

L38: Kriging Surrogate Models

Simulation Without a Black Box

Outline

Surrogate Model Concept

Gaussian Process Regression

Latin Hypercube Sampling

Fitting a Kriging Model

Expected Improvement

Bayesian Optimization Loop

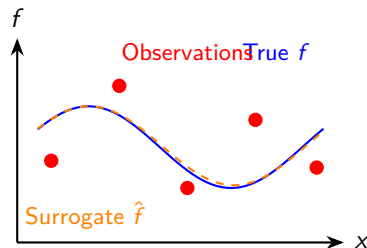
When Surrogates Help and When They Don't

Key Takeaways

Surrogate Models: Cheap Approximation to Expensive Simulation

Core Idea

Run the simulation at n design points $\{x_1, \dots, x_n\}$, observe noisy outputs $y_i = f(x_i, \omega_i)$. Fit a *surrogate* $\hat{f}(x)$ that is cheap to evaluate. Use $\hat{f}$ to screen the decision space before running more expensive simulations.



When is a surrogate useful?

- Each simulation run takes ≥ 1 minute
- Decision space is continuous or large
- Response surface is smooth

Gaussian Process Regression

GP Prior

A **Gaussian process** is a distribution over functions:

$$f(x) \sim \mathcal{GP}(\mu(x), k(x, x'))$$

- $\mu(x)$: prior mean (often 0)
- $k(x, x')$: covariance (kernel) — encodes smoothness

After observing data $\mathbf{y}$ at $\mathbf{X}$:

$$f(x^*) \mid \mathbf{X}, \mathbf{y} \sim \mathcal{N}(\mu^*(x^*), \sigma^{*2}(x^*))$$

Prediction is a *mean estimate* plus *uncertainty*.

Popular kernels:

- **RBF (squared exponential):**

$$k(x, x') = \sigma^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right)$$

Infinitely differentiable, smooth.

- **Matérn 5/2:** less smooth, often better in practice for simulation outputs.

Kriging = GP Regression

In engineering, “Kriging” refers to the same GP posterior mean predictor, originating in geostatistics.

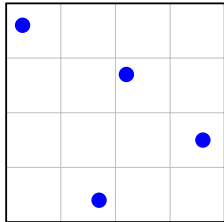
Design of Experiments: Latin Hypercube Sampling

LHS Definition

Divide each input dimension into n equal strata.

Sample one point from each stratum so that no two points share a stratum in *any* dimension.

Result: n points with better space-filling than simple random sampling.



LHS, $n = 4$, $d = 2$

```
from scipy.stats.qmc import LatinHypercube
import numpy as np
```

```
sampler = LatinHypercube(d=2, seed=0)
sample = sampler.random(n=20)
# scale to problem bounds
lb = np.array([5, 20])
ub = np.array([30, 100])
X = lb + sample * (ub - lb)
# X[:,0]: reorder point s
# X[:,1]: order-up-to level S
```

Fitting Kriging with scikit-learn

```

from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import Matern, WhiteKernel

```

```

kernel = Matern(nu=2.5) + WhiteKernel(noise_level=0.1)
gp = GaussianProcessRegressor(kernel=kernel,
                              n_restarts_optimizer=10,
                              random_state=42)

```

```

# X_train: LHS design points (n_samples x n_features)
# y_train: mean simulation output at each point
gp.fit(X_train, y_train)

```

```

# Predict with uncertainty at new points
y_pred, y_std = gp.predict(X_test, return_std=True)
# y_pred: posterior mean | y_std: posterior std dev

```

Noise Level

WhiteKernel absorbs simulation noise (heteroskedastic). If runs are replicated, use the empirical variance per design point instead.

Expected Improvement Acquisition Function

Expected Improvement (EI)

Let $f^+ = \min_i y_i$ (best observed so far). At a candidate point x :

$$\text{EI}(x) = \mathbb{E}[\max(f^+ - f(x), 0)]$$

For a GP posterior with mean μ^* and std σ^* :

$$\text{EI}(x) = (f^+ - \mu^*) \Phi(Z) + \sigma^* \phi(Z)$$

where $Z = (f^+ - \mu^*)/\sigma^*$, Φ and ϕ are the standard normal CDF and PDF.

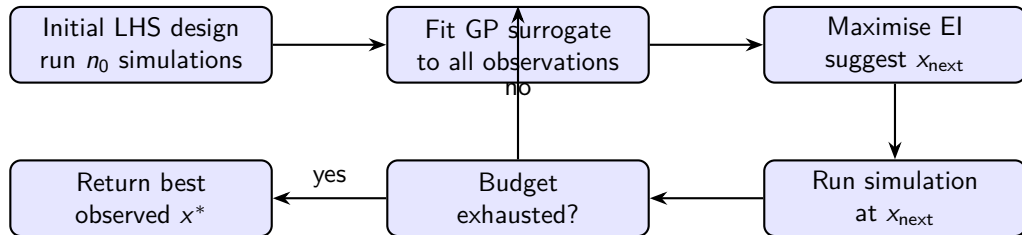
Exploration–exploitation tradeoff:

- High μ^* (low predicted cost) $\Rightarrow$ exploitation
- High σ^* (uncertain region) $\Rightarrow$ exploration
- EI balances both automatically

Next evaluation point:

$$x_{\text{next}} = \arg \max_{x \in \mathcal{X}} \text{EI}(x)$$

Bayesian Optimization Loop



Each iteration adds exactly one simulation evaluation and refits the GP.

When Surrogates Help and When They Don't

Surrogates help when:

- Each simulation run is expensive ($\geq$ minutes)
- Response surface is smooth (few discontinuities)
- Dimensionality $d \leq 10$
- A training budget of $\geq 5d$ runs is available

Surrogates struggle when:

- $d \gg 10$ (curse of dimensionality in GP)
- Response is highly multi-modal or discontinuous
- Simulation noise is very large relative to signal
- The decision space is purely discrete with no structure

Warning

A surrogate predicts the *mean* response — it cannot replace statistical output analysis. Always revalidate the surrogate optimum with fresh simulation replications and a proper CI.

Key Takeaways

L38 Summary

1. A surrogate fits a cheap approximation to the expensive simulation response surface, enabling efficient search.
2. Gaussian Process regression provides a mean prediction *and* uncertainty estimate — both are used in optimization.
3. Latin Hypercube Sampling is the preferred design for initial surrogate training ($\geq 5d$ points recommended).
4. Expected Improvement balances exploitation (low predicted cost) and exploration (high uncertainty) automatically.
5. Surrogates are most effective for smooth, low-dimensional ($d \leq 10$), expensive simulation problems.

Next: L39 — MDP framing for simulation control.